# Macroeconomics I — MPE Insper

Practice list: Golden rule and the two-period competitive equilibrium (Prova 01, Q2)

**Instructions.** Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

**Notation (Kurlat 2020).** Capital share $\alpha$ in $Y = AK^{\alpha}L^{1-\alpha}$ (the exam wrote it $\sigma$); $\sigma$ is reserved for the curvature of $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ ($1/\sigma$ is the intertemporal elasticity of substitution, $\sigma = 1$ is $\ln c$); $\beta \in (0,1)$ discount factor, $\rho \equiv 1/\beta - 1$; $\delta$ depreciation; $n$ population growth; $a$ assets carried into the next period; $r^K$ rental rate of capital, $r$ interest rate, with $1 + r = r^K + 1 - \delta$ (Kurlat (9.3.11)). Lower-case $k$, $y$, $c$ are per worker unless stated. Book exercises are reproduced verbatim, in the book's own notation, after a source line; parts this list adds are marked.

**What this list repairs.** (i) Golden-rule algebra: carry the exponent $\frac{\alpha}{1-\alpha}$, cross-check with MPK $= n + \delta$, test a knife edge. (ii) The household problem: every choice variable and every income source, corners included. (iii) Market clearing turns one household's saving rule into a law of motion $k_{t+1} = \phi(k_t)$; the saving rate it implies is chosen, and need not be the golden-rule rate.

## Block 1 — Exam level: the golden rule and the two-period household

*(Kurlat §§4.3, 6.2; Jones ch. 5 • 35 points)*

---

**Question 1.** (8 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics, *Exercise 5.8 "National Accounts and The Golden Rule", pp. 97–98 — statement reproduced verbatim, all items (a)–(d). What this trains:* reading $\alpha$, $s$ and $\delta$ off the national accounts, then deciding on which side of the golden rule the economy sits, by two independent routes (the saving rate against $\alpha$; the MPK against $n + \delta$).

In 2018, the following events happened in Aurum, which is a closed economy that uses the Denarius (plural: Denarii) as its currency.

- Grano, Inc. hired Cornelia to plant and harvest 500 tons of wheat on a large plot of land using a tractor. Both the land and the tractor are owned by Grano, Inc.
  - It paid Cornelia 200 Denarii for her work on the harvest.
  - It sold the wheat to Panem, Inc. for 1,000 Denarii.
- The tractor the Cornelia used:
  - Was new at the beginning of the year. It was worth 2,000 Denarii.
  - Was no longer new at the end of the year. By then it was worth 1,800 Denarii.
- Panem, Inc. hired Gaius to grind the wheat in its mill and produce a million loaves of bread.
  - It paid Gaius 1,400 Denarii for his work.
  - It sold the bread for 4,000 Denarii to the hungry citizens, who eat it.
- Fabrica, Inc. hired Livia to build a combine harvester.
  - It paid Livia 400 Denarii for her work.
  - It sold the combine harvester to Grano, Inc. for 1,000 Denarii.

(a) Construct GDP accounts (in Denarii) for Aurum by production, income and expenditure.

(b) What were the income shares of labor and capital?

(c) What was the savings rate?

(d) Assume the following:

- the savings rate is constant,
- the production function has the form:

$$Y = AK^{\alpha}L^{1-\alpha}$$

and there is no technological progress,

- the depreciation rate for the tractor is typical for the capital stock as a whole,
- the population is constant.

In the long run, would the level of consumption be higher if the economy slightly increased its saving rate?

*Solution:*

**Question 2.** (10 points) *Source: Jones, C. I. (2020),* Macroeconomics*, 5th ed., ch. 5, Review Question 4, p. 129 — statement reproduced verbatim; item (a) asks it in Jones's model with the exponent* $1/3$*; items (b)–(d) are added by this list. What this trains:* the steady-state consumption formula with the right exponent, and the exam's exact algebra slip.

**Review Question 4 (Jones).** What is the solution of the Solow model for consumption per person in the steady state, $c^* \equiv C^*/L^*$? How does each parameter in the solution affect $c^*$, and why?

**a)** (2 points) Answer Review Question 4 in Jones's model ($Y = \bar{A}K^{1/3}\bar{L}^{2/3}$, $\Delta K = sY - \bar{d}K$).

*Added parts (not in the book):*

**b)** (2 points) Find the $s$ that maximises $c^*$. Check it with the MPK condition.

**c)** (3 points) In the exam, a student substituted $k^*$ instead of $k^{*\alpha}$ and maximised $(1-s)\left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}}$. For a general capital share $\alpha$, which saving rate does that give? Compare it with the true golden rule as $\alpha \to 0$ and explain which one passes this knife-edge test, and why.

**d)** (3 points) Let $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$, $K_{t+1} = (1-\delta)K_t + sY_t$. With $\tilde{k} \equiv K/(AL)$, derive $\tilde{k}_{t+1}(\tilde{k}_t)$, the steady state $\tilde{k}^*$, consumption per effective worker $\hat{c}^*(s)$ and the golden-rule $s$. Show that at any steady state MPK $= \alpha N/s$, with $N \equiv (1+n)(1+g) - 1 + \delta$. Evaluate $\hat{c}^*$ at $s \in \{0.2,\ s_{GR},\ 0.6\}$ for $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$.

*Solution:*

**Question 3.** (10 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 6.1 "Two-Period Problem with Taxes and Initial Wealth", pp. 121–122 — statement reproduced verbatim, all items (a)–(e). What this trains:* the full household problem with every income source, closed forms, and how $\sigma$ decides whether $c_1$ rises or falls with $r$ (the log case $\sigma = 1$ is the exam's).

Suppose a household solves the following two-period consumption-savings problem with taxes:

$$\max_{c_1,a,c_2} \; u(c_1) + \beta u(c_2)$$

$$\text{s.t.}$$

$$a = a_0 + y_1 - \tau_1 - c_1$$

$$c_2 = y_2 - \tau_2 + (1+r)\,a$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, where: $c_1$ is consumption at time 1, $c_2$ is consumption at time 2, $y_1$ is household income at time 1, $y_2$ is household income at time 2, $\tau_1$ are taxes at time 1, $\tau_2$ are taxes at time 2, and $a_0$ is initial wealth.

(a) Solve for the household's choice of $c_1$, $c_2$ and $a$ in closed form.[1]

(b) How does $\frac{c_1}{y_1}$ depend on $y_2$? What would happen if households suddenly became optimistic about the future?

(c) How does $\frac{c_1}{y_1}$ depend on $a_0$? Interpret.

(d) How does $\frac{c_1}{y_1}$ depend on $\beta$? Interpret.

(e) Suppose $y_2 = \tau_1 = \tau_2 = 0$ and compute $\frac{\partial c_1}{\partial r}$. How does the answer depend on $\sigma$? Interpret the answer. [Hint: this is a hard question, not the maths but the interpretation. Think about what $\sigma$ means for the relative importance of income and substitution effects]

*Solution:*

**Question 4.** (7 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 6.2 "Savings Rates for Different People", p. 122 — statement reproduced verbatim, all items (a)–(b). Its Exercise 6.1 is Question 3 of this list. What this trains:* a saving rate out of *current* income depends on the income profile and, through human wealth, on $r$, even when the saving rate out of *lifetime* wealth does not.

An economist has data on the occupation, this year's income (denoted $y$) and this year's consumption (denoted $c$) of a sample of 26-year-olds. Within this sample, some are top professional athletes and others are medical doctors in their first year of residence. [Make whatever assumptions you think are reasonable and, if you wish, refer back to Exercise 6.1]

(a) Suppose one computed $s = \frac{y-c}{y}$ for each individual in the sample. Should we expect $s$ to be higher for athletes or doctors?

(b) Suppose interest rates go down. How should we expect the response of $s$ to differ between the two groups?

*Solution:*

[1] Kurlat's footnote: "In closed form" means that you have an explicit expression for something. Suppose I ask you to determine some variable $x$. Then something like $x = y^2 - z$ is a solution "in closed form"; something like $e^x + x - b = 0$ also implicitly tells what $x$ should be but is not in closed form.

## Block 2 — Harder: equilibrium, endogenous saving, and the golden rule

*(Kurlat §§9.1–9.3; Romer §2.9–2.11 • 37 points)*

---

**Question 5.** (6 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 9.2 "Storage", pp. 179–180 — statement reproduced verbatim, all items (a)–(c). What this trains:* prices come from the firm's FOCs and the no-arbitrage condition $1 + r = r^K + 1 - \delta$; the Euler equation then signs the consumption path.
Suppose the production function is the following:

$$F(K, L) = K$$

and the rate of depreciation is $\delta = 1$.

(a) Why is this production function called "storage"?

(b) What will be the real interest rate and the wage in this economy?

(c) With standard preferences, will consumption increase over time, remain constant over time or decrease over time?

*Solution:*

**Question 6.** (12 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercises 9.11 "Patience and Investment", p. 185, and 9.12 "Optimal vs Fixed Savings Rates", pp. 185–186 (the key synthesis) — statements reproduced verbatim, all items. Change: 9.12 is worked in Python (the book points to Matlab code); item (f) is added by this list. What this trains:* the saving rate Solow assumes is the one the optimising economy chooses in the long run; it is below $s_{GR}$.
*Added (not in the book): work 9.11 symbolically, and 9.12 in Python (the book points to Matlab code).*

**9.11 Patience and Investment**

Suppose the production function is $F(K, L) = K^\alpha L^{1-\alpha}$, and the labor supply is exogenous and equal to $L = 1$. The representative household's preferences are given by:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

and the depreciation rate is $\delta$.

(a) Use equation (9.3.17) to find an expression for the level of capital that this economy will have in the steady state.

(b) Find an expression for the level of GDP that the economy will have in the steady state.

(c) Find an expression for the steady-state investment-to-GDP ratio. How does it depend on $\beta$? Explain.

**9.12 Optimal vs Fixed Savings Rates**

Suppose the economy is as in Exercise 9.11, with $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$ and $\sigma = 2$. The economy starts with an initial capital stock $K_0 = 2$. We are going to compare how this economy behaves under a fixed investment rate and under an optimally-chosen investment rate.

(a) First suppose that, like we assumed in Chapter 4, the investment rate is exogenous and constant, and equal to the steady-state investment-to-GDP ratio that you found in Exercise 9.11. Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.

(b) Now suppose that consumption and investment are chosen in accordance with (9.3.13). Find the level of initial consumption that is consistent with the system of equations (9.3.15) - (9.3.13) and the economy converging to a steady state. (You can find a Matlab code that will compute this at the book website). Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.

(c) Plot the consumption paths of the two economies over the first 150 years on the same graph.

(d) Replace the computed paths of consumption from parts (a) and (b) into the utility function to compute how much utility the representative household will obtain in the first 150 years of each economy.

(e) By what fraction would one have to reduce consumption in the optimal-savings economy for the household to get the same utility as in the exogenous-savings economy?

*Added parts (not in the book):*

**(f)** Compare the steady-state investment rate with the golden-rule rate and the steady-state capital with $K_{GR}$.

*Solution:*

**Question 7.** (9 points) *Source: Romer, D. (2012),* Advanced Macroeconomics*, 4th ed., Problem 2.16 "Depreciation in the Diamond model and microeconomic foundations for the Solow model", p. 97 — statement reproduced verbatim, all items (a)–(b), preceded by the set-up and part (a) of Problem 2.15, p. 97, which 2.16(b) refers to (2.15(b)–(d) omitted); item (c) is added by this list. Harder, contrast. What this trains:* the exam's "Solow with $\delta = 1$" remark done right, in the overlapping-generations reading where the young save only their wage.

*Added (not in the book): Problem 2.15 is included because 2.16(b) refers to it.*

**2.15.** *A discrete-time version of the Solow model.* Suppose $Y_t = F(K_t, A_t L_t)$, with $F(\bullet)$ having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1 + g)A_t$, $L_{t+1} = (1 + n)L_t$, and $K_{t+1} = K_t + sY_t - \delta K_t$.

(a) Find an expression for $k_{t+1}$ as a function of $k_t$.

**2.16.** *Depreciation in the Diamond model and microeconomic foundations for the Solow model.* Suppose that in the Diamond model capital depreciates at rate $\delta$, so that $r_t = f'(k_t) - \delta$.

(a) How, if at all, does this change in the model affect equation (2.59) giving $k_{t+1}$ as a function of $k_t$?

(b) In the special case of logarithmic utility, Cobb–Douglas production, and $\delta = 1$, what is the equation for $k_{t+1}$ as a function of $k_t$? Compare this with the analogous expression for the discrete-time version of the Solow model with $\delta = 1$ from part (a) of Problem 2.15.

*Added parts (not in the book):*

**(c)** Compare the implied saving rate with $s_{GR}$ and state when this economy over-accumulates.
*Solution:*

**Question 8.** (10 points) *Conceptual, words only.* True or false? Justify each in at most four lines, without algebra.

**a)** (2 points) In the exam's economy (log utility over $c_1, c_2$; one unit of time; work carries no disutility; $w_1 > 0$), no first-order condition for $L_1$ exists, so $L_1$ is left undetermined.

**b)** (2 points) With $u = \ln c$ the saving rate does not depend on $r$; so in Kurlat 6.2 a rate cut leaves every person's $s = (y - c)/y$ unchanged.

**c)** (2 points) The First Welfare Theorem holds in Kurlat's infinite-horizon economy, so its steady state is the golden rule.

**d)** (2 points) An economy with capital above the golden rule is dynamically inefficient: a permanently lower saving rate raises consumption at every date, starting today.

**e)** (2 points) Once the household's rule $a_1 = \frac{\beta}{1+\beta} \times$(resources) is known, the economy's law of motion $k_2 = \phi(k_1)$ follows without further conditions.

## Block 3 — Created: the exam's economy, extended

*(Kurlat §§6.2, 7.1, 9.1, 4.3; Romer §2.9–2.11 • 28 points)*

---

**Question 9.** (15 points) *The exam's household with leisure, inside an economy.* Periods $t = 1, 2, \ldots$ Each period a generation of size one is born and lives two periods, with one unit of time when young and none when old. Generation $t$ maximises $\ln c_{1t} + \psi \ln(1 - L_t) + \beta \ln c_{2t+1}$, $\psi > 0$, $0 < \beta < 1$. Technology $Y_t = AK_t^{\alpha} L_t^{1-\alpha}$, $\delta = 1$, competitive markets; saving is lent to firms as next period's capital and earns $1 + r_{t+1} = r_{t+1}^K$. *Generation 1 is the exam's household:* it is born with assets $a_0 \geq 0$ and owns the firm, which owns $K_1 > 0$, so in period 1 it receives the profit $\Pi_1 = Y_1 - w_1 L_1$ besides its wage. From period 2 on, the old rent their capital to firms, which earn zero profit; each young generation earns only its wage.

**a)** (2 points) Write generation 1's two period budgets and its present-value budget constraint. List every choice variable and every income source, and read the constraint in words.

**b)** (4 points) Derive all first-order conditions and closed forms for $c_1, c_2, L_1, a_1$ given $(w_1, \Pi_1, r_2, a_0)$. When is $L_1$ interior? Show that $a_1$ is the share $\frac{\beta}{1+\beta}$ of the resources $a_0 + w_1 L_1 + \Pi_1$. What happens as $\psi \to 0$?

**c)** (3 points) Let $a_0 = 0$. Using the firm's FOC, find equilibrium $L_1^*$ (show it depends on neither $K_1$ nor $A$) and the map $K_2 = \phi_1(K_1)$.

**d)** (3 points) For $t \geq 2$ find the young's labour $\bar{L}$, the law of motion $K_{t+1} = \phi(K_t)$, the steady state and its stability. Draw $\phi$ against the 45° line for $\alpha = 1/3$, $\beta = 0.8$, $\psi = 0.5$, $A = 5$, $K_1 = 1$, marking $K^*$ and the path from $K_2$.

**e)** (3 points) Compare with Solow ($\delta$, $n$, $s$) and with $s_{GR}$. When does this economy over-accumulate? Evaluate for $(\alpha, \beta) = (1/3, 0.8)$ and $(0.2, 0.9)$, with the interest rate in each. Why does generation 1 work less than later ones, and what goes wrong if $\Pi_1$ is left out of its budget?

*Solution:*

**Question 10.** (13 points) *CRRA and population growth: when the saving rate depends on $r$.* Overlapping generations as in Question 9 but without leisure: each young person supplies one unit of labour, earns only the wage, and the number of young grows, $L_{t+1} = (1+n)L_t$. Utility $u(c_{1t}) + \beta u(c_{2t+1})$ with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$; $y = Ak^\alpha$ per young worker; $\delta = 1$.

**a)** (3 points) Solve the young's problem. Show that saving is a share $s(r)$ of the wage, find $s(r)$, and sign $s'(r)$. Explain the sign through income and substitution effects.

**b)** (3 points) Using the firm's FOCs and market clearing $K_{t+1} = L_t a_t$, write the law of motion for $k \equiv K/L$. Show that $k_{t+1}$ is unique given $k_t$ when $\sigma \leq 1$, and give it in closed form for $\sigma = 1$.

**c)** (3 points) Show that the steady state satisfies $s(r)(1+r) = \frac{\alpha(1+n)}{1-\alpha}$ and that the aggregate saving rate is $\frac{I}{Y} = \frac{\alpha(1+n)}{1+r}$. Conclude: $\frac{I}{Y} > s_{GR} \iff r < n \iff k^* > k_{GR}$.

**d)** (2 points) Let $\alpha = 0.2$, $\beta = 0.5$, $n = 0.3$, $A = 5$. For $\sigma = 1$ find $r^*$, $k^*$, $k_{GR}$ and steady-state consumption per worker at both. For $\sigma \in \{1/2, 2\}$ find $r^*$ numerically (Python) and say which economies are dynamically inefficient, and why $\sigma$ matters.

**e)** (2 points) Draw the $\sigma = 1$ map against the 45° line, marking $k^*$ and $k_{GR}$. In what sense is this "Solow with $\delta = 1$"?

*Solution:*